

COMPILER DESIGN LAB 6

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1. <https://godbolt.org/>
2. In LHS select language as C, RHS select compiler as x86-64 clang 20.1.0 and next to it in compiler options type -O0 -emit-llvm
3. In LHS paste any C code like shown above and output will be shown in RHS

A. C CODE 1

```
#include <stdio.h>

int main() {
    for(int i=0; i<5; i++) {
        for(int j=0; j<5; j++) {
            printf("*");
        }
        printf("\n");
    }
    return 0;
}
```

INPUT:

← → ↺ ⚙️ godbolt.org

COMPILER EXPLORER Add... ▾ More ▾ Templates

Look after yourself, and, if you can, so

C source #1

Save/Load + Add new... ▾ Vim C

```

1 #include <stdio.h>
2 int main() {
3     for(int i=0; i<5; i++) {
4         for(int j=0; j<5; j++) {
5             printf("");
6         }
7         printf("\n");
8     }
9     return 0;
10 }
```

OUTPUT:

FINAL SS:

[illegible]

B. C CODE 2

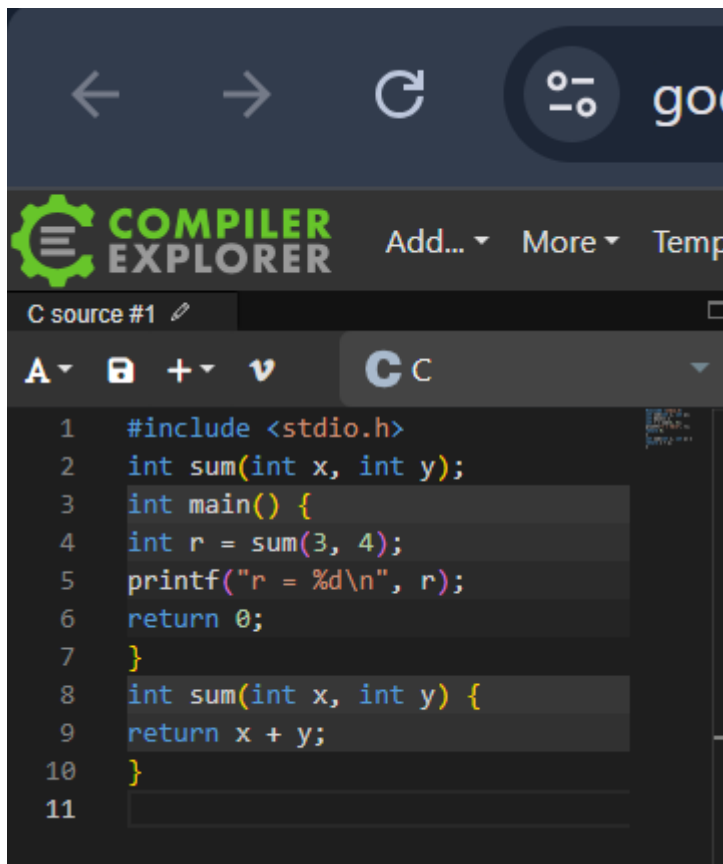
```
#include <stdio.h>

int sum(int x, int y);

int main() {
    int r = sum(3, 4);
    printf("r = %d\n", r);
    return 0;
}

int sum(int x, int y) {
    return x + y;
}
```

INPUT:

A screenshot of the Compiler Explorer web application. The browser's address bar shows a Google search bar. The Compiler Explorer logo is in the top left, with navigation links 'Add...', 'More', and 'Temp'. The file name 'C source #1' is displayed. The code editor shows the same C code as the previous block, with line numbers 1 through 11 on the left. The code is:

```
1  #include <stdio.h>
2  int sum(int x, int y);
3  int main() {
4  int r = sum(3, 4);
5  printf("r = %d\n", r);
6  return 0;
7  }
8  int sum(int x, int y) {
9  return x + y;
10 }
11
```

OUTPUT:

```
1 @.str = private unnamed_addr constant [8 x i8] c"r - 5d/0A/00", align 1, !dbg 10
2
3 define @main() @.main() #0 !dbg 118 {
4     %1 = alloca i32, align 4
5     %2 = alloca i32, align 4
6     store i32 0, ptr %1, align 4
7     @dbg_declare(ptr %2, i32, !dbgexpression(), !24)
8     %3 = call i32 @llvm.i32.roundf.f, i32 roundf 4, !dbg 125
9     store i32 %3, ptr %2, align 4, !dbg 126
10    %4 = load i32, ptr %2, align 4, !dbg 126
11    %5 = call i32 @llvm.i32.roundf.f, i32 roundf 4, !dbg 127
12    ret i32 0, !dbg 128
13 }
14
15 define @main() @.main() @.main() #0 !dbg 129 {
16     %1 = alloca i32, align 4
17     %2 = alloca i32, align 4
18     store i32 0, ptr %1, align 4
19     @dbg_declare(ptr %2, i32, !dbgexpression(), !13)
20     store i32 %1, ptr %2, align 4
21     @dbg_declare(ptr %3, i32, !dbgexpression(), !135)
22     %3 = load i32, ptr %3, align 4, !dbg 136
23     %4 = load i32, ptr %4, align 4, !dbg 137
24     %5 = add nsw i32 %3, %4, !dbg 138
25     ret i32 %5, !dbg 139
26 }
27
28 declare i32 @printf(ptr roundf, ...) #1
29
30 attributes #0 = { noinline nounwind optnone uwtable "frame-pointer"="all" "min-legal-vector-width"="0" "no-trapping-math"="true" "stack-protector-buffer-size"="8" "target-cpu"="x86-64" "target-features"="+cmov,ecx,efsr,mmx,ssse3,x87" "tune-cpu"="generic" }
31 attributes #1 = { "frame-pointer"="all" "no-trapping-math"="true" "stack-protector-buffer-size"="8" "target-cpu"="x86-64" "target-features"="+cmov,ecx,efsr,mmx,ssse3,x87" "tune-cpu"="generic" }
```

FINAL SS:

```
1 @.str = private unnamed_addr constant [8 x i8] c"r - 5d/0A/00", align 1, !dbg 10
2
3 define @main() @.main() #0 !dbg 118 {
4     %1 = alloca i32, align 4
5     %2 = alloca i32, align 4
6     store i32 0, ptr %1, align 4
7     @dbg_declare(ptr %2, i32, !dbgexpression(), !24)
8     %3 = call i32 @llvm.i32.roundf.f, i32 roundf 4, !dbg 125
9     store i32 %3, ptr %2, align 4, !dbg 126
10    %4 = load i32, ptr %2, align 4, !dbg 126
11    %5 = call i32 @llvm.i32.roundf.f, i32 roundf 4, !dbg 127
12    ret i32 0, !dbg 128
13 }
14
15 define @main() @.main() @.main() #0 !dbg 129 {
16     %1 = alloca i32, align 4
17     %2 = alloca i32, align 4
18     store i32 0, ptr %1, align 4
19     @dbg_declare(ptr %2, i32, !dbgexpression(), !13)
20     store i32 %1, ptr %2, align 4
21     @dbg_declare(ptr %3, i32, !dbgexpression(), !135)
22     %3 = load i32, ptr %3, align 4, !dbg 136
23     %4 = load i32, ptr %4, align 4, !dbg 137
24     %5 = add nsw i32 %3, %4, !dbg 138
25     ret i32 %5, !dbg 139
26 }
27
28 declare i32 @printf(ptr roundf, ...) #1
29
30 attributes #0 = { noinline nounwind optnone uwtable "frame-pointer"="all" "min-legal-vector-width"="0" "no-trapping-math"="true" "stack-protector-buffer-size"="8" "target-cpu"="x86-64" "target-features"="+cmov,ecx,efsr,mmx,ssse3,x87" "tune-cpu"="generic" }
31 attributes #1 = { "frame-pointer"="all" "no-trapping-math"="true" "stack-protector-buffer-size"="8" "target-cpu"="x86-64" "target-features"="+cmov,ecx,efsr,mmx,ssse3,x87" "tune-cpu"="generic" }
```

C. C CODE 3

```
// word.c

#include <stdio.h>

#include <stdlib.h>

#include <string.h>

char *replaceWord(const char *s, const char *oldW, const char *newW) {

char *result;

int i, cnt = 0;

int newWlen = strlen(newW);

int oldWlen = strlen(oldW);
```

```

for (i = 0; s[i] != '\0'; i++) {
    if (strstr(&s[i], oldW) == &s[i]) {
        cnt++;
        i += oldWlen - 1;
    }
}

result = (char *)malloc(i + cnt * (newWlen - oldWlen) + 1);

i = 0;

while (*s) {
    if (strstr(s, oldW) == s) {
        strcpy(&result[i], newW);
        i += newWlen;
        s += oldWlen;
    } else {
        result[i++] = *s++;
    }
}

result[i] = '\0';

return result;
}

// Driver Program

int main() {
    char str[] = "A jellyfish stinged me on the beach.";
    char c[] = "stinged";

    char d[] = "stung";
    char *result = NULL;

    printf("Old string: %s\n", str);
    result = replaceWord(str, c, d);

```

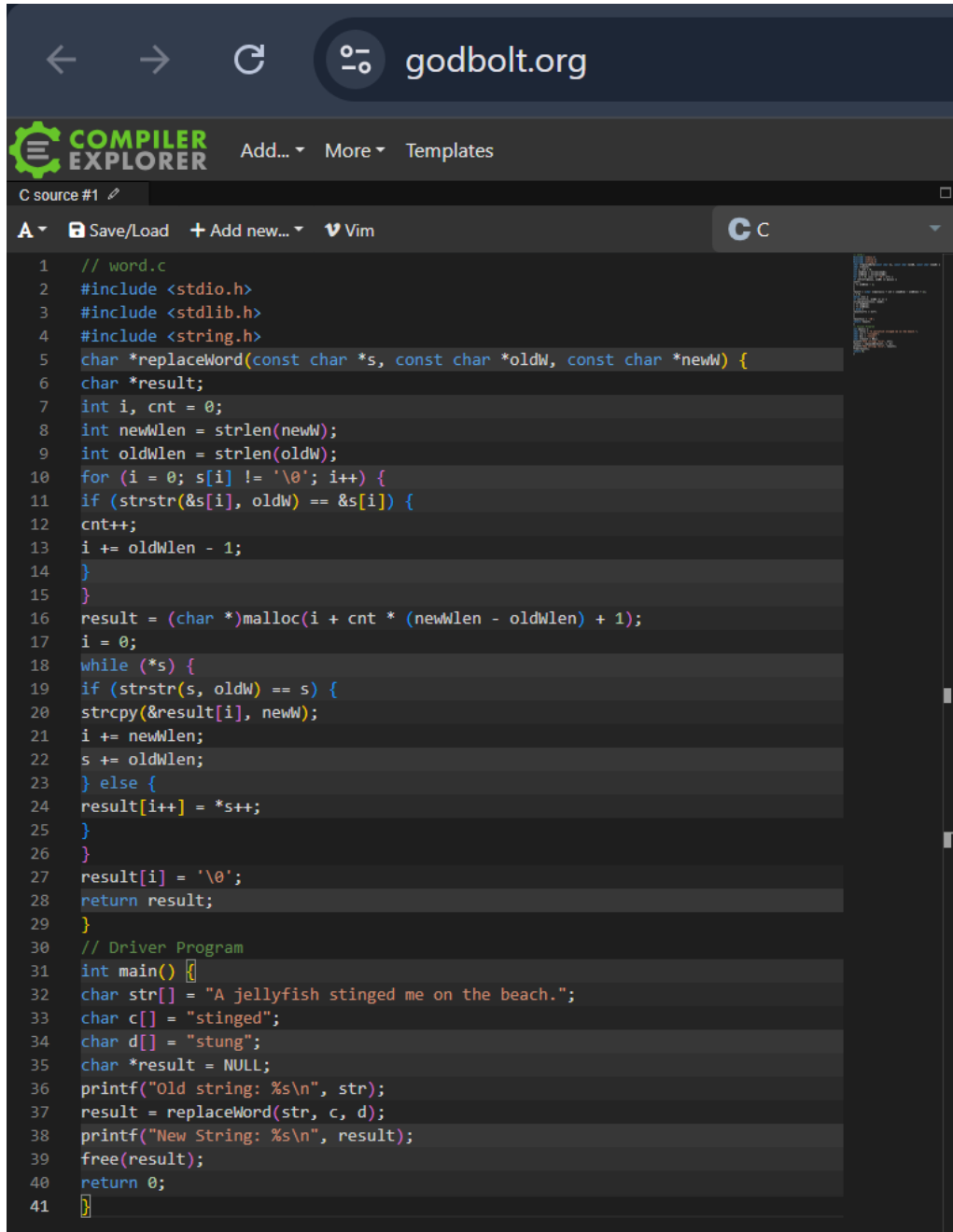
```
printf("New String: %s\n", result);

free(result);

return 0;

}
```

INPUT:

The image shows a screenshot of the Godbolt Compiler Explorer web application. The browser's address bar displays 'godbolt.org'. The application header includes the 'COMPILER EXPLORER' logo and navigation links like 'Add...', 'More', and 'Templates'. Below the header, the editor is titled 'C source #1'. The main area contains a C program with line numbers 1 through 41. The program defines a 'replaceWord' function that replaces occurrences of 'oldW' with 'newW' in a string 's'. It uses 'strstr' to find occurrences and 'strcpy' to copy the new word. The 'main' function demonstrates this by replacing 'stinged' with 'stung' in the sentence 'A jellyfish stinged me on the beach.'. The code is syntax-highlighted, and the interface has a dark theme.

```
1 // word.c
2 #include <stdio.h>
3 #include <stdlib.h>
4 #include <string.h>
5 char *replaceWord(const char *s, const char *oldW, const char *newW) {
6     char *result;
7     int i, cnt = 0;
8     int newWlen = strlen(newW);
9     int oldWlen = strlen(oldW);
10    for (i = 0; s[i] != '\0'; i++) {
11        if (strstr(&s[i], oldW) == &s[i]) {
12            cnt++;
13            i += oldWlen - 1;
14        }
15    }
16    result = (char *)malloc(i + cnt * (newWlen - oldWlen) + 1);
17    i = 0;
18    while (*s) {
19        if (strstr(s, oldW) == s) {
20            strcpy(&result[i], newW);
21            i += newWlen;
22            s += oldWlen;
23        } else {
24            result[i++] = *s++;
25        }
26    }
27    result[i] = '\0';
28    return result;
29 }
30 // Driver Program
31 int main() {
32     char str[] = "A jellyfish stinged me on the beach.";
33     char c[] = "stinged";
34     char d[] = "stung";
35     char *result = NULL;
36     printf("Old string: %s\n", str);
37     result = replaceWord(str, c, d);
38     printf("New String: %s\n", result);
39     free(result);
40     return 0;
41 }
```

OUTPUT:

[illegible]

```
121 X79 = add maw 112 X78, X77, fdmg 1185
122 store 112 X79, ptr X7, align 4, fdmg 1185
123 X80 = load 112, ptr X71, align 4, fdmg 1185
124 X81 = load ptr, ptr X4, align 8, fdmg 1187
125 X82 = sext 112 X80 to 164, fdmg 1187
126 X83 = getelementptr inbounds i8, ptr X81, 164 X82, fdmg 1187
127 store ptr X81, ptr X4, align 8, fdmg 1187
128 br label X73, fdmg 1188
129
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131
132 X85 = load ptr, ptr X4, align 8, fdmg 1189
133 X86 = getelementptr inbounds maw i8, ptr X85, 112 1, fdmg 1189
134 store ptr X86, ptr X4, align 8, fdmg 1189
135 X87 = load i8, ptr X86, align 1, fdmg 1111
136 X88 = load ptr, ptr X7, align 8, fdmg 1112
137 X89 = load 112, ptr X5, align 8, fdmg 1113
138 X90 = add maw 112 X89, 1, fdmg 1113
139 store 112 X90, ptr X2, align 4, fdmg 1113
140 X91 = sext 112 X90 to 164, fdmg 1113
141 X92 = getelementptr inbounds i8, ptr X88, 164 X91, fdmg 1112
142 store i8 X92, ptr X82, align 1, fdmg 1114
143 br label X83
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```

D. C CODE 4

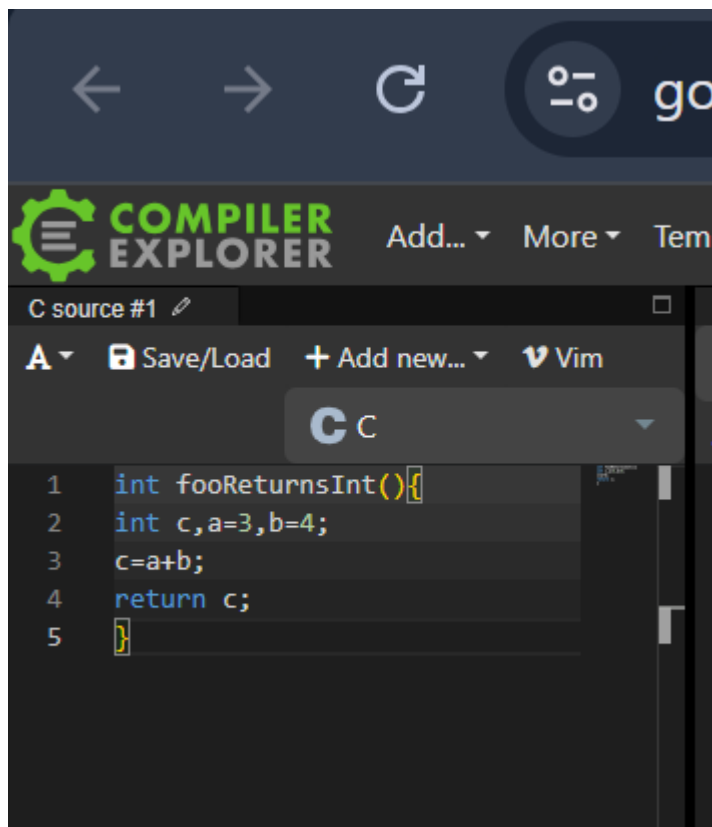
```
int fooReturnsInt(){
int c,a=3,b=4;

c=a+b;

return c;

}
```

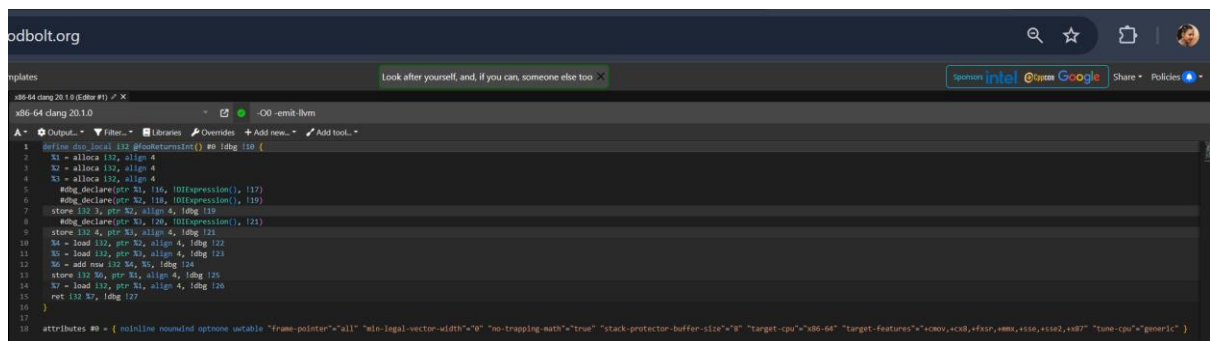

INPUT:



The screenshot shows the Compiler Explorer interface. The top bar includes navigation icons and the 'go' button. The 'COMPILER EXPLORER' logo is on the left, and 'Add...', 'More', and 'Temp' buttons are on the right. Below the logo, the text 'C source #1' is visible. A toolbar contains 'A', 'Save/Load', '+ Add new...', and 'Vim'. A dropdown menu shows 'C' selected. The code editor contains the following C code:

```
1 int fooReturnsInt(){
2   int c,a=3,b=4;
3   c=a+b;
4   return c;
5 }
```

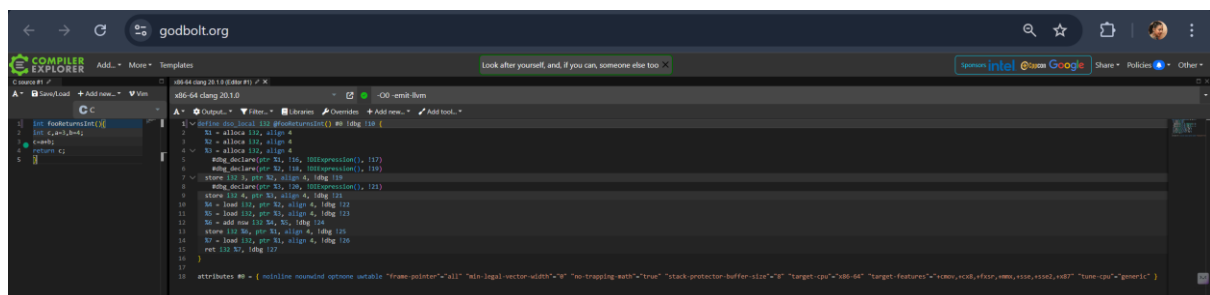
OUTPUT:



The screenshot shows the Compiler Explorer interface with the assembly output. The top bar includes navigation icons and the 'go' button. The 'COMPILER EXPLORER' logo is on the left, and 'Add...', 'More', and 'Temp' buttons are on the right. Below the logo, the text 'C source #1' is visible. A toolbar contains 'A', 'Save/Load', '+ Add new...', and 'Vim'. A dropdown menu shows 'C' selected. The code editor contains the following assembly code:

```
1 define __local_132 @fooReturnsInt() #0 fdng 110 {
2   %1 = alloca i32, align 4
3   %2 = alloca i32, align 4
4   %3 = alloca i32, align 4
5   @dng.declare(ptr %1, 116, 101Expression(), 117)
6   @dng.declare(ptr %2, 118, 101Expression(), 119)
7   store i32 3, ptr %2, align 4, fdng 119
8   @dng.declare(ptr %3, 120, 101Expression(), 121)
9   store i32 4, ptr %3, align 4, fdng 121
10  %4 = load i32, ptr %2, align 4, fdng 122
11  %5 = load i32, ptr %3, align 4, fdng 123
12  %6 = add nsw i32 %4, %5, fdng 124
13  store i32 %6, ptr %1, align 4, fdng 125
14  %7 = load i32, ptr %1, align 4, fdng 126
15  ret i32 %7, fdng 127
16 }
17
18 attributes #0 = [ noinline nounwind optnone uwtable "frame-pointer"="all" "min-legal-vector-width"="0" "no-trapping-math"="true" "stack-protector-buffer-size"="0" "target-cpu"="x86-64" "target-features"="+cmov,ssse3,fxsr,sse2,x87" "tune-cpu"="generic" ]
```

FINAL SS:



The screenshot shows the Compiler Explorer interface with the final assembly output. The top bar includes navigation icons and the 'go' button. The 'COMPILER EXPLORER' logo is on the left, and 'Add...', 'More', and 'Temp' buttons are on the right. Below the logo, the text 'C source #1' is visible. A toolbar contains 'A', 'Save/Load', '+ Add new...', and 'Vim'. A dropdown menu shows 'C' selected. The code editor contains the following assembly code:

```
1 define __local_132 @fooReturnsInt() #0 fdng 110 {
2   %1 = alloca i32, align 4
3   %2 = alloca i32, align 4
4   %3 = alloca i32, align 4
5   @dng.declare(ptr %1, 116, 101Expression(), 117)
6   @dng.declare(ptr %2, 118, 101Expression(), 119)
7   store i32 3, ptr %2, align 4, fdng 119
8   @dng.declare(ptr %3, 120, 101Expression(), 121)
9   store i32 4, ptr %3, align 4, fdng 121
10  %4 = load i32, ptr %2, align 4, fdng 122
11  %5 = load i32, ptr %3, align 4, fdng 123
12  %6 = add nsw i32 %4, %5, fdng 124
13  store i32 %6, ptr %1, align 4, fdng 125
14  %7 = load i32, ptr %1, align 4, fdng 126
15  ret i32 %7, fdng 127
16 }
17
18 attributes #0 = [ noinline nounwind optnone uwtable "frame-pointer"="all" "min-legal-vector-width"="0" "no-trapping-math"="true" "stack-protector-buffer-size"="0" "target-cpu"="x86-64" "target-features"="+cmov,ssse3,fxsr,sse2,x87" "tune-cpu"="generic" ]
```